Concept map: Models: regression, classification and gradient descent

Normal distribution
Independent and mutually exclusive events
Bayes' theorem
Probability density function (PDF)
Naïve Bayes
What deep learning is
Types of neural networks
Cosine similarity
XGBoost
Linear and quadratic programming
Kernel trick
Overfitting
Feature selection
Vector magnitude, distance and scalar operations
KNN imputer
Dot product
Regularisation
SMOTE
Lasso regression
Lagrange multipliers, KKT and duality
Cross-validation
Equation of a hyperplane
Classification problems
Support vector machines
Bias-variance trade-off
Curse of dimensionality
Polynomial features
Polynomial regression
Elastic Net
Instance-based learning
K-nearest neighbours
Singular value decomposition
Perceptron trick
Hinge loss and soft margin
Ridge regression
Moore-Penrose pseudo-inverse
Convex sets and convex optimisation
Standardization
Perceptron
Feature scaling
Linear combinations, span and basis
Logistic regression
Model-based learning
Multicollinearity
Random forest
Softmax regression
Jacobian and matrix gradients
Balanced random forest
Unbalanced data
Normal equation
Problem with the perceptron (XOR)
Multiple linear regression
One-hot encoding
Cost-sensitive learning
ANN for classification
Perceptron loss
Assumptions of linear regression
Log loss (binary cross entropy)
Regression problems
Gradient descent
Q-Q plot
Convex and non-convex loss
Confusion matrix
Simple linear regression
Entropy, information gain and Gini
Categorical and sparse categorical cross-entropy
Gradient boosting
Batch gradient descent
Multi-layer perceptron (MLP)
ANN for regression
Iterative imputation (MICE)
Sigmoid function
Ordinary least squares (closed form)
Decision trees
Keras workflow
Best-fit line and squared error
Stochastic gradient descent
Neural networks
Backpropagation
Derivatives of one variable
Supervised learning
Learning rate
Universal approximation theorem
Partial derivatives and gradients
Loss functions in deep learning
Regression trees
Mini-batch gradient descent
Regression metrics
Hessian and multivariate Taylor
MLP notation and parameter count
Batch size in Keras
Forward propagation
Covariance and covariance matrix
Memoization
Vanishing gradient
Normalization
Online learning
Huber loss
Matrix multiplication as composition
Training curves (History)
ReLU
Exploding gradient and gradient clipping

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